KRAFZEN AI – Finalized MVP Tech Stack

Guiding Principle

The MVP demonstrates AI-driven CAD automation with minimal cost and complexity. It must be easy to deploy, reliable for demos, and anticipated cost less than $100/month to operate.

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1. Frontend

Category	Stack	Reason	Cost
Framework	**Next.js (React)**	Modern, performant, and deploys instantly on Vercel.	Free
Styling	**Tailwind CSS**	Lightweight, responsive, and developer-friendly.	Free
State Mgmt	**React Context API**	Built-in, no external dependency for MVP.	Free
Deployment	**Vercel**	Easiest for CI/CD and SSL. Great free tier.	Free (Pro optional $20/mo)

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2. Backend

Category	Stack	Reason	Cost
Framework	**FastAPI (Python)**	Already integrated with Whisper and ezdxf; async and simple.	Free
Hosting	**Render.com**	Seamless Python support, simple Git deploys.	Free (upgrade ~$7–20/mo)

Why Render, not Vercel: Vercel's backend functions are serverless (10s timeout, no file persistence). Whisper and DXF generation need persistent runtime and local file access. Render provides that without extra setup.

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3. Database & Storage

Category	Stack	Reason	Cost
Database	**Supabase (PostgreSQL)**	Managed Postgres + API + built-in Auth; easiest setup.	Free (up to 500MB DB)
Authentication	**Supabase Auth (JWT)**	Handles users and sessions without custom backend code.	Free
File Storage	**Supabase Storage**	Direct uploads for DXF, JSON, and previews.	Free (2GB tier)

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4. AI Services

Category	Stack	Reason	Cost
Transcription	**OpenAI Whisper API**	High accuracy, minimal setup, pay-per-use.	~$0.006/minute
Parsing / LLM	**GPT-4 Turbo API**	Reliable JSON output for geometry parsing.	~$0.03 / 1K tokens
Estimated Usage	~5 hrs audio + 100 GPT calls	**$40–60/month**	

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5. CAD Generation

Category	Stack	Reason	Cost
CAD Engine	**ezdxf (Python)**	Free, open-source DXF generation library.	Free
Viewer	**Canvas-based DXF preview**	Displays 2D preview from generated file.	Free
Advanced (Later)	Autodesk Forge (optional)	Add only when scaling to Revit integration.	~$100–200/mo if used

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6. Infrastructure & Deployment

Category	Stack	Reason	Cost
CI/CD	**GitHub Actions**	Automates build & deploy to Vercel/Render.	Free (2k min/month)
Containerization	**Docker Compose**	Optional for local demo consistency.	Free
Domain & SSL	**Cloudflare**	Free SSL + DNS.	Free

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7. Analytics & Monitoring

Category	Stack	Reason	Cost
Error Tracking	**Sentry (Free Tier)**	Catch runtime errors during demo.	Free
Usage Analytics	**Google Analytics 4**	Track pageviews and engagement.	Free
PostHog	**Free Tier**	Heatmaps / seeing how users interact	Free (paid later)

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8. Estimated Monthly Costs

Category	Approx. Monthly Cost
OpenAI APIs (Whisper + GPT-4 Turbo)	$40–60
Render Backend Hosting	$0–20
Supabase DB & Storage	$0–25
Total Monthly MVP Cost	**≈ $40–105**

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9. Architecture Overview

Frontend (Next.js + Tailwind)
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Backend (FastAPI)

/transcribe > Whisper API
/parse > GPT-4 Turbo
/generate > ezdxf → DXF

Supabase (DB + Auth + Storage)

Deployment:

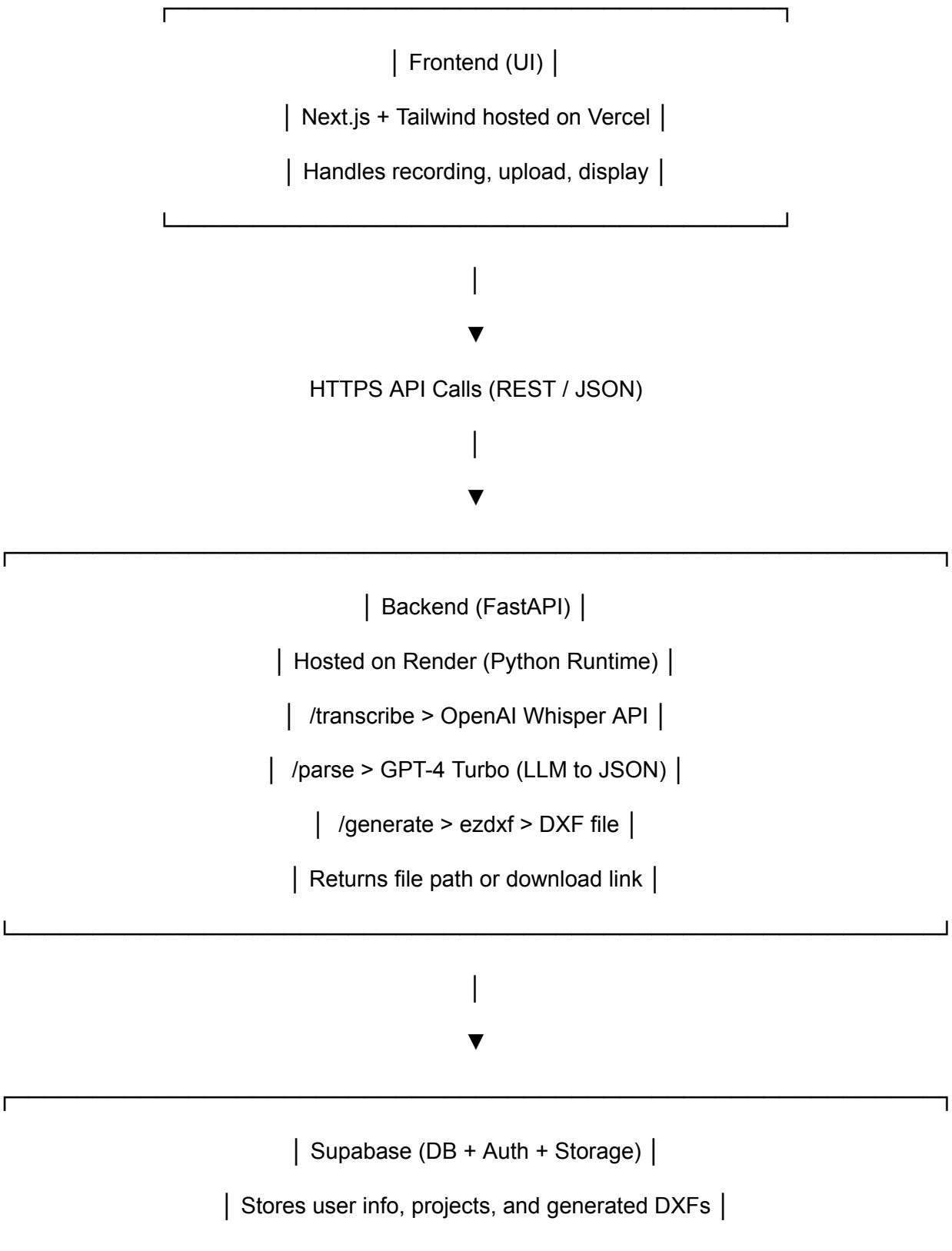
- **Frontend:** Vercel (Git auto-deploy)
- **Backend:** Render (FastAPI build)
- **Database & Storage:** Supabase
- **Domain:** Cloudflare

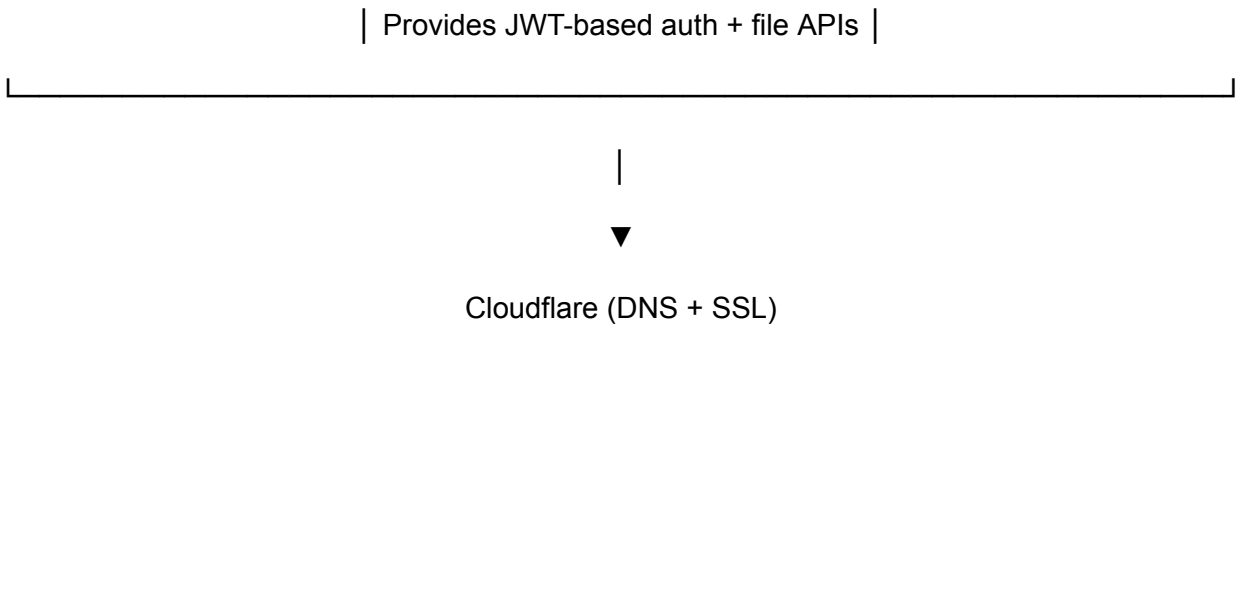
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10. Why This Stack

- **All-in-one simplicity:** FastAPI + Supabase + Next.js covers everything.
- **No GPU or ops overhead:** Fully API-driven.
- **Free hosting tiers:** Everything runs approx $75/mo.
- **Fully demoable:** Real AI > DXF workflow works live.
- **Evolvable:** Seamlessly extend to Revit/Forge later.

Summary:
This is the finalized, cost-optimized MVP tech stack for KRAFZEN AI. It's lean, modern, and focused on delivering a functional, investor-ready demonstration without DevOps complexity or recurring infrastructure overhead.


Frontend (UI)
Next.js + Tailwind hosted on Vercel
Handles recording, upload, display
HTTPS API Calls (REST / JSON)
Backend (FastAPI)
Hosted on Render (Python Runtime)
/transcribe > OpenAI Whisper API
/parse > GPT-4 Turbo (LLM to JSON)
/generate > ezdxf > DXF file
Returns file path or download link
Supabase (DB + Auth + Storage)
Stores user info, projects, and generated DXFs


Provides JWT-based auth + file APIs
Cloudflare (DNS + SSL)